Signal Processing: Methods and Algorithms
Laboratory 8

1 Introduction

A random process $x[n]$ has two possible behaviors according to the hypotheses

$$H_0 \quad : \quad x[n] = \xi[n] \tag{1}$$
$$H_1 \quad : \quad x[n] = A + \xi[n] \tag{2}$$

where $n = 0, \ldots, N-1$, $\xi[n]$ is a white Gaussian noise with zero mean and variance σ^2, and A is a positive constant. The corresponding Neyman-Pearson (NP) detector which maximizes the probability of detection P_D for a given probability of false alarm P_{FA} decides H_1 when

$$\hat{\mu}_x > \gamma \tag{3}$$

where γ is the threshold and $\hat{\mu}_x$ is the sample mean estimator

$$\hat{\mu}_x = \frac{1}{N} \sum_{n=0}^{N-1} x[n] \tag{4}$$

For a given P_{FA}, it can be shown that

$$\gamma = \sqrt{\frac{\sigma^2}{N}} Q^{-1}(P_{FA}) \tag{5}$$
$$P_D = Q\left(\frac{\gamma - A}{\sqrt{\sigma^2/N}}\right) \tag{6}$$

where Q is the right-tail cumulative distribution function defined as

$$Q(z) = \int_z^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt \tag{7}$$

and Q^{-1} is its inverse. The Q and Q^{-1} functions are given in the Matlab functions `Q.m` and `Qinv.m`, respectively.

2 Exercise 1: Theoretical detector performances as a function of the mean value

Generate the receiver operating characteristics (ROCs) of the NP detector described in Sect. 1 for the P_{FA} values

$$P_{FA} = \mathsf{0.01 : 0.08 : 0.97} \tag{8}$$

and for the amplitude values

$$A = 0.1,\ 0.5,\ 1,\ 2,\ 4 \tag{9}$$

with $N = 1$ and $\sigma = 1$. **Hint:** start by generating the ROC for one value of the amplitude A, for example $A = 1$.

3 Exercise 2: Theoretical detector performances as a function of the number of samples

Generate the ROCs of the NP detector described in Sect. 1 for the P_{FA} values

$$P_{FA} = \mathsf{0.01 : 0.08 : 0.97} \tag{10}$$

and for the number of samples

$$N = 1,\ 2,\ 4,\ 16 \tag{11}$$

with $A = 1$ and $\sigma = 1$.

4 Exercise 3: Simulated detector performances

Estimate by numerical simulations the ROCs of the NP detector described in Sect. 1 with $N = 2$, $A = 1$, and $\sigma = 1$ for the P_{FA} values

$$P_{FA} = \mathsf{0.01 : 0.08 : 0.97} \tag{12}$$

Then obtain the theoretical ROCs. Finally, compare the estimated and theoretical ROCs in the same picture. Use a solid line for the theoretical ROCs and a circular marker (option `'o'`, lowercase letter "o" in the Matlab function `plot`) for the estimated ones.

Hint: Obtain the values of the threshold γ from Eq. (5) for the P_{FA} values in Eq. (12). Then, for each γ value, estimate P_{FA} by doing the following:

1. Simulate $M = 10^4$ realizations (or more) of $x[n]$ under the H_0 hypothesis
2. Count the number of detections N_D occurring
3. Estimate P_{FA} as

$$\hat{P}_{FA} = \frac{N_D}{M} \tag{13}$$

To obtain $\hat{P}_D$ repeat the same steps by simulating the H_1 hypothesis.